

个人作业报告

题目：**编译原理**

理论课第一次作业

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【摘　要】

随着元宇宙、数字孪生和 VR/AR 等应用的兴起[1], 同时具备高保真视觉重建与物理一致性仿真的统一三维表示成为计算机图形学与计算机视觉交叉领域的重要课题。近年来兴起的 3D 高斯泼溅 (3D Gaussian Splatting, 3DGS) 以离散各向异性高斯作为渲染基元, 在效率与质量之间取得了良好平衡, 为构建统一的 “渲染-仿真” 框架提供了新契机。本文围绕 Gaussian-Flow、PhysGaussian 和 PhysDreamer 三项代表性工作, 系统梳理了基于 3DGS 的动态 4D 重建与物理仿真研究进展。首先, 我们介绍 Gaussian-Flow 如何通过双域变形模型与自适应时间缩放, 在不依赖深度网络的前提下实现高效、紧凑的 4D 场景表示。其次, 我们分析 PhysGaussian 将 3D 高斯直接视为物质点法 (MPM) 中的物理粒子, 在统一管线中完成从连续介质力学到可微渲染的映射, 并通过内部填充与各向异性正则化获得稳定且多样的材质行为。然后, 我们讨论 PhysDreamer 如何利用视频生成模型蕴含的世界先验, 通过可微仿真与蒸馏优化反推物体的空间材质场, 从而赋予静态 3D 资产以感知合理的交互动态。最后, 文章从精度目标、驱动机制和计算代价等维度对三者进行综合对比, 展望物理基础模型与实时数字孪生等未来发展方向。

关键词： 3D 高斯泼溅, 4D 重建, 物理仿真, Gaussian-Flow, PhysGaussian, PhysDreamer

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1 Exercise 1.1

1.1 Problem

1) Imagine an artificial computer language, which can be utilized to solve a practical problem, i.e. the application of the language.
   - Tips 1. Language is an alternative approach to problem solving.
   - Tips 2. First find a proper problem, then design a language to solve the problem.
2) Give an example of a complete piece written in the proposed language.
3) Discuss how to define the new language and try your approach.
4) Describe the process of changing the thinking of your language to a reality, i.e. how to make the artificial language usable.

1.2 Solution

1.2.1 Problem 1

我设计的语言想要解决的问题是实现计算器的某些功能，包括加、减、乘、除、幂、对数和开根号。这种语言所解决的计算均为二元计算，每条语句以运算符开始，后面为两个待计算的数字，以分号结束，并且运算符与两个操作数以“&”分开，中间可以有随意数量的空格。运算符（假设第一个数字为 a，第二个为 b）：

- `+-*/` 分别代表加减乘除；
- `^` 代表幂运算，后面第一个为底数，第二个数字为指数，即 a^b；
- `$` 代表对数运算，第一个数为底数，第二个数为真数，即 $\log_a b$；
- `@` 表示开根号，第一个数字为根指数，第二个数为准备开根号的数字，即 $\sqrt[a]{b}$。

1.2.2 Problem 2

```
+ & 3 & 5;    //3+5
* & 2 & 3;    //2*3
/ & 10 & 5;   //10/5
^ & 2 & 3;    //2 的 3 次方
```

```
$ & 2 & 8;   //以2为低8的对数
@ & 2 & 4;   //开根号，4的2次方跟，即根号2
```

1.2.3 Problem 3

对语言的定义如下:

- 字符集:

{1,2,3,4,5,6,7,8,9,0,+,−,∗,/,^,$,@,;}

- BNF:

```
<计算语句> ::= <运算符> & integer & integer ;
<运算符> ::= + | − | ∗ | / | ^ | $ | @
integer ::= [ symbol ] unsigned
unsigned ::= digit { digit }
symbol ::= + | −
digit ::= 0 | 1 | ... | 9
```

1.2.4 Problem 4

如果想要实现这门语言，当读到指令之后按照 & 进行分词，先读第一个操作符，针对每种计算分别编写不同的程序，然后读后面的两个操作数，再传给程序计算即可。

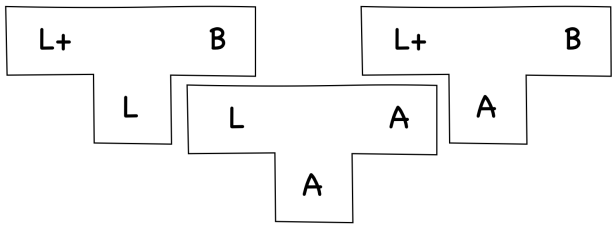
2 Exercise 1.2

2.1 Problem

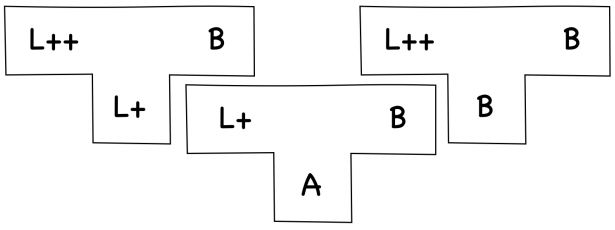
1) Draw a T-diagram with two stages of bootstrappings.
   - Given a new programming language L++, we firstly implement L, a small subset of L++.
   - Then we use L to implement L+, a subset of L++ and a superset of L.
   - Finally, L++ is implemented using L+.

2.2 Solution

Stage 1



Stage 2



参考文献

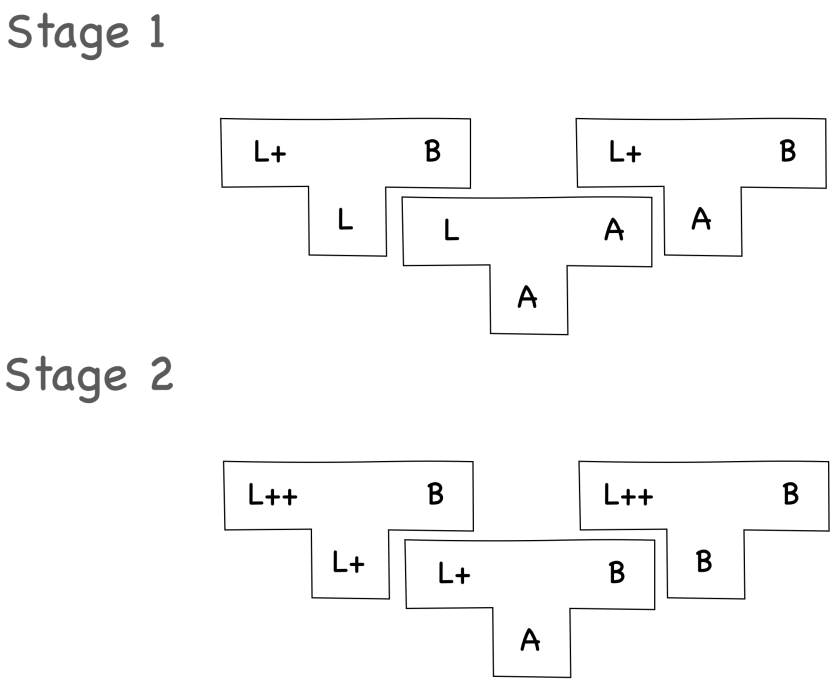
[1] JUN LI W. Lecture01. introduction: Principles of compiler construction[Z]. 2026.

附录 A 实验细节

A.1 实验截图

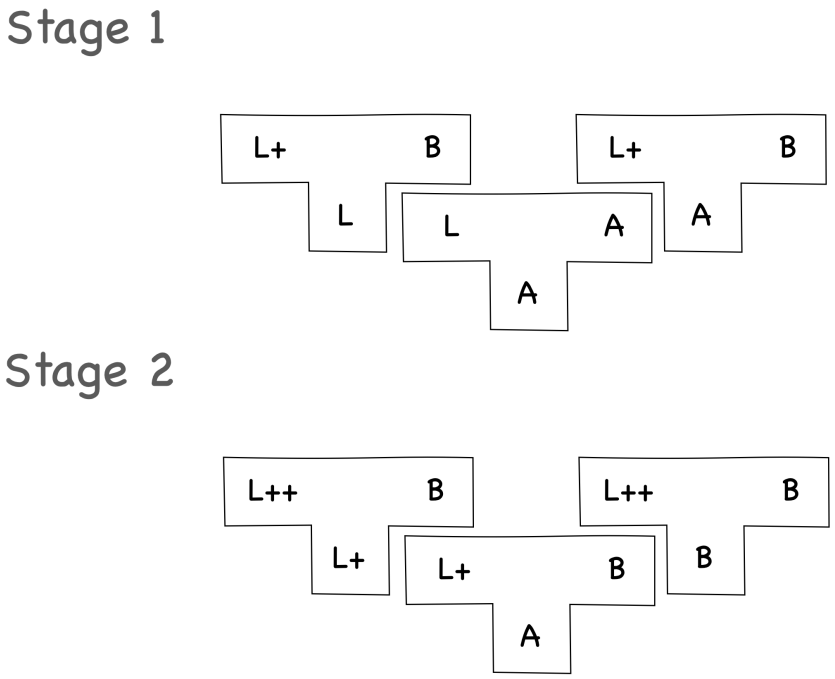
A.1.1 截图 a 示例

截图 a 说明了 aaa



A.1.2 截图 b 示例

截图 b 说明了 bbb



A.2 代码

核心算法代码展示

A.2.1 算法 A 代码

Listing A.1 算法 A: demo.cpp

```cpp
#include <iostream>
int main()
{
    ::std::cout << "hello world" << ::std::endl;
    return 0;
}
```

A.2.2 算法 B 代码

Listing A.2 算法 B: demo.cpp

```cpp
#include <iostream>
int main()
{
    ::std::cout << "hello world" << ::std::endl;
    return 0;
```

```
}
```

A.3 Use of AI

This section documents the interactions between the author and AI assistant during the development of this MCM/ICM paper. The conversations are presented in a question-and-answer format to demonstrate how AI was utilized for various aspects of the research.

A.3.1 Model Development Assistance

Question: How to optimize the UAV path planning algorithm?

I'm working on a UAV path planning algorithm for wildfire monitoring. The current approach uses a grid-based method, but I'm encountering issues with computational efficiency when scaling to larger areas. What optimization techniques would you recommend?

AI Response

For UAV path planning optimization in wildfire scenarios, consider these approaches:

1. **Hybrid algorithms**: Combine genetic algorithms with simulated annealing for better convergence 2. **Multi-resolution grids**: Use coarse grids for initial planning, then refine locally 3. **Dynamic programming**: Implement Dijkstra's algorithm with priority queues 4. **Parallel processing**: Divide the search space for concurrent computation

The key is to balance exploration and exploitation while considering real-time constraints.

A.3.2 Data Processing and Analysis

Question: Data preprocessing for wildfire detection

I have satellite imagery data for wildfire detection, but there's significant noise and missing values. What preprocessing steps should I implement before applying machine learning models?

AI Response

Recommended preprocessing pipeline:

- **Missing data**: Use interpolation or k-nearest neighbors imputation
- **Noise reduction**: Apply Gaussian or median filters
- **Normalization**: Scale pixel values to [0,1] range
- **Feature extraction**: Consider texture features, thermal gradients
- **Data augmentation**: Rotate, flip, and scale images for robustness

Also, implement cross-validation to ensure model generalization.

A.3.3 Conclusion on AI Usage

The AI assistant provided valuable support in algorithm optimization, data preprocessing guidance, mathematical formulation, and code implementation. All final decisions, analysis, and interpretations were made by the human author, with AI serving as a collaborative tool for brainstorming and technical guidance.